



DATA ANALYTICS PORTFOLIO PROJECT

Airbnb Dynamic Pricing Recommendation Engine

Benchmarking every NYC listing's price against comparable market medians to guide data-driven pricing decisions.

 Python

 Excel

 Power BI

48,895 NYC Listings | 5 Boroughs | Median-Benchmark Pricing Logic

Business Problem & Objective



The Problem

Airbnb hosts frequently price listings by intuition rather than by comparing against similar, nearby listings — leaving money on the table or pricing themselves out of bookings.

41.1% of listings priced below comparable market median

38.8% priced above the median — a competitiveness risk

Source: engineered pricing_status field, 48,895 listings



The Objective

Design a data-driven recommendation engine that benchmarks each listing's current price against the median price of comparable listings — same borough and same room type — and returns a clear pricing action.

- 1 Compare Price
- 2 Benchmark vs. Median
- 3 Recommend Action

Dataset & Tools



New York City Airbnb Listings Dataset

48,895

Listings

16

Raw Columns

5

Boroughs

221

Neighborhoods

3

Room Types

0

Duplicate Rows

Data Quality

last_review / reviews_per_month: 20.6% missing (10,052 rows) — no review history yet

name / host_name: < 0.1% missing (26 / 22 rows)

Duplicate records: 0 — dataset verified clean

Analytics Workflow



Python

Data cleaning, EDA & feature logic — Pandas, NumPy, Matplotlib



Excel

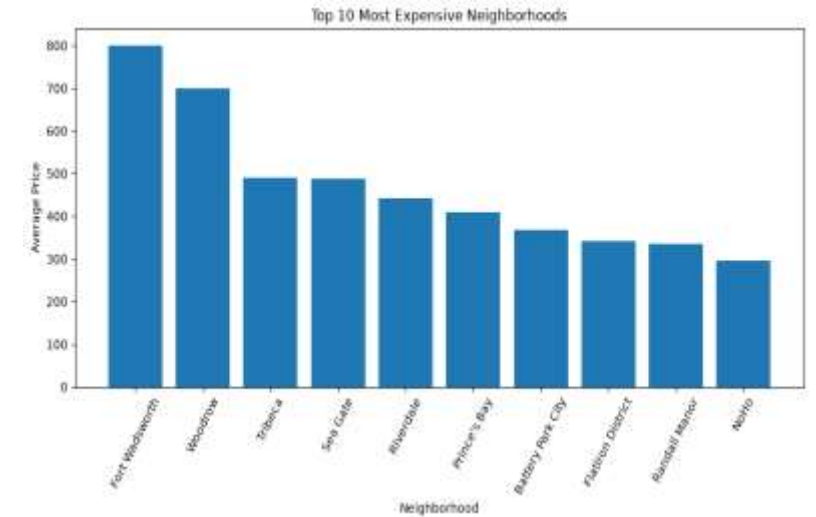
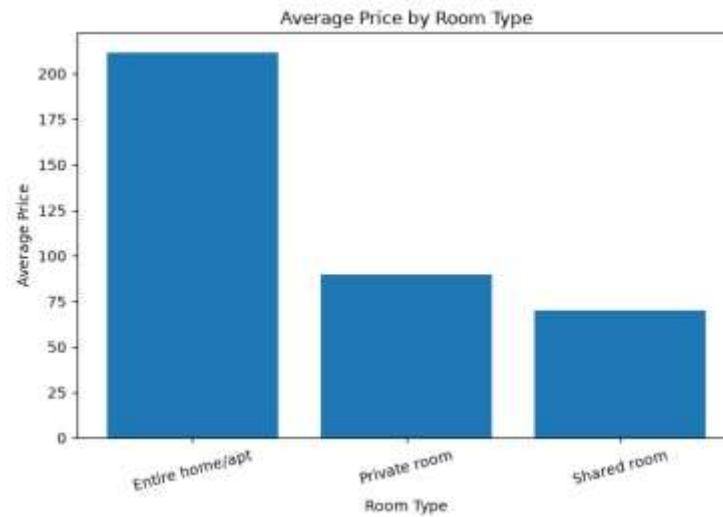
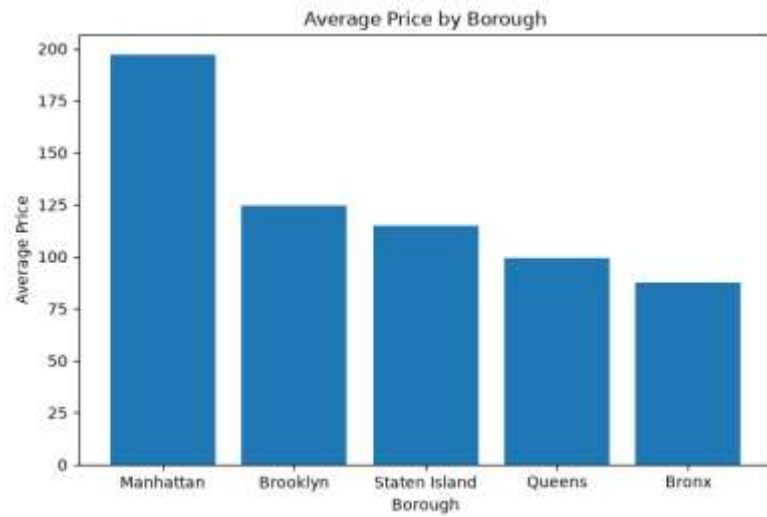
Engineered categories & the pricing recommendation columns



Power BI

Interactive 3-page dashboard for stakeholder storytelling

Exploratory Data Analysis — Key Signals



What the Data Shows

- Manhattan averages \$196.88/night vs. \$87.50 in the Bronx — a 2.25x price gap across boroughs.
- Entire home/apt averages \$211.79/night, nearly 3x a shared room's \$70.13 — room type drives price more than any other factor.
- The 10 priciest neighborhoods (e.g. Fort Wadsworth \$800, Tribeca \$490.64) sit well above the citywide median of \$106.

Feature Creation for the Pricing Engine



price_category

Budget / Standard / Premium / Luxury tiers from price

23,917 Budget · 16,583 Standard · 7,340 Premium · 1,044 Luxury



availability_category

Not Available / Low / Medium / High from availability_365

17,533 · 13,060 · 6,686 · 11,616 listings respectively



review_category

Low / Medium / High / Very High from number_of_reviews

30,323 · 11,613 · 3,964 · 2,995 listings respectively

The Three Columns That Power the Recommendation Engine

recommended_price

Median price of comparable listings (same borough + room type)

price_difference

recommended_price minus the listing's current price

pricing_status

Increase / Maintain / Lower — the final recommendation label

How the Recommendation Engine Works

Each listing is compared only to its true peer group — listings in the same borough (neighbourhood_group) and of the same room_type. The median price of that peer group becomes the fair-market benchmark.

```
recommended_price = median( price ) grouped by
[neighbourhood_group, room_type]
```

Peer-Group Median Price Benchmark (\$)

Borough	Entire home/apt	Private room	Shared room
Manhattan	\$191	\$90	\$69
Brooklyn	\$145	\$65	\$36
Queens	\$120	\$60	\$37
Bronx	\$100	\$53.5	\$40
Staten Island	\$100	\$50	\$30

Real medians computed from the engineered dataset (airbnb_EDA_Engineered.xlsx)

Decision Logic

 price < 90% of benchmark
→ **Increase Price**

 within ±10% of benchmark
→ **Maintain Price**

 price > 110% of benchmark
→ **Consider Lowering**

Median chosen over mean to stay resistant to extreme outlier prices (e.g. the \$10,000 max listing) within each peer group.

Recommendation Rules & Outcomes



Increase Price

`price < 90% of recommended_price`

20,104

41.1% of listings



Maintain Price

`within ±10% of recommended_price`

9,808

20.1% of listings



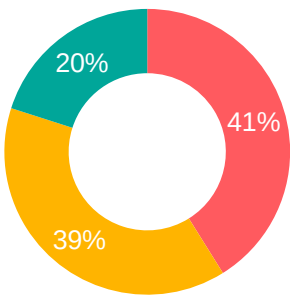
Consider Lowering

`price > 110% of recommended_price`

18,983

38.8% of listings

Distribution of Pricing Recommendations



- Increase Price
- Consider Lowering
- Maintain Price

Reading the Result

Nearly 4 in 5 listings sit outside the $\pm 10\%$ comfort zone — most hosts are leaving revenue on the table (underpriced) or risking lower booking rates (overpriced). Only 1 in 5 listings is already priced optimally against comparable inventory.

Interactive Pricing Intelligence



Dynamic Pricing Recommendation Engine — page 3 of 3



Page 1 — Market Overview



Page 2 — Market & Pricing Insights

SO WHAT?

Business Insights



Borough price gap

Manhattan averages \$196.88/night — 2.25x Bronx's \$87.50 — the single biggest price driver in the market.



Demand vs. price

Brooklyn logs more total reviews (486,574) than pricier Manhattan (454,569), suggesting stronger booking turnover at lower price points.



Room type premium

Entire home/apt commands \$211.79 on average — 2.4x a private room (\$89.78) and 3x a shared room (\$70.13).



Revenue opportunity

41.1% of listings (20,104) are priced below their peer-group median — the largest single group in the recommendation output.



Supply vs. turnover

Staten Island has the highest average availability (199.7 of 365 days) — a signal of lower demand relative to booking pace.

Limitations & Conclusion



Limitations

- Peer group is defined only by borough + room type — amenities, review quality, and seasonality aren't factored in.
- Raw data includes listings priced at \$0, a likely data artifact not fully filtered before analysis.
- Median can be less stable in small segments (e.g. Staten Island shared rooms, $n < 40$).
- Single static snapshot — no time-series or seasonal demand adjustment.
- Delivered as a transparent, rule-based benchmark rather than a predictive ML pricing model.



Conclusion

An end-to-end analytics pipeline — Python cleaning & EDA, Excel feature engineering, a transparent rule-based pricing engine, and a Power BI dashboard — turns 48,895 raw listings into a clear, explainable action for every host: increase, maintain, or lower price.



Data cleaning & EDA



Feature engineering



Business rule design



Dashboard storytelling